PostgreSQL Data Types Cheat Sheet

Quick reference for the most useful data types and when to use each

1. Numeric

Type	Size	Range / Notes	Use for
`SMALLINT`	2 bytes	-32,768 to 32,767	Small counts, flags
`INTEGER` / `INT`	4 bytes	about ±2.1 billion	General whole numbers
`BIGINT`	8 bytes	about ±9.2 × 10^18	Large IDs, big counters
`NUMERIC(p,s)` / `DECIMAL(p,s)`	variable	Exact, user-defined precision	Money, exact values
`REAL`	4 bytes	about 6 digits precision	Approximate decimals
`DOUBLE PRECISION`	8 bytes	about 15 digits precision	Scientific values
`SERIAL` / `BIGSERIAL`	4 / 8 bytes	Auto-increment	Primary keys (older style)
`GENERATED ALWAYS AS IDENTITY`	4 / 8 bytes	Auto-increment	Primary keys (modern, preferred)

2. Character / String

Type	Description
`CHAR(n)`	Fixed length, padded with spaces (avoid)
`VARCHAR(n)`	Variable length, max n characters
`TEXT`	Variable length, no limit (fine for most cases)

3. Date and Time

Type	Example	Notes
`DATE`	`2026-10-28`	Date only
`TIME`	`14:30:00`	Time only
`TIMESTAMP`	`2026-10-28 14:30:00`	No time zone
`TIMESTAMPTZ`	`2026-10-28 14:30:00+05:30`	With time zone (preferred)
`INTERVAL`	`3 days 2 hours`	Duration

4. Boolean

Type	Values
`BOOLEAN`	`TRUE`, `FALSE`, `NULL`

5. Binary

Type	Use for
`BYTEA`	Raw binary data (small files, hashes)

6. JSON

Type	Notes
JSON	Stores exact text, slower to query
JSONB	Binary, indexable, faster (preferred)

```
SELECT extra_info->>'city'
FROM student
WHERE extra_info @> '{"active": true}';
```

7. UUID

Type	Example
UUID	a0eebc99-9c0b-4ef8-bb6d-6bb9bd380a11

```
id UUID DEFAULT gen_random_uuid() PRIMARY KEY
```

8. Array

Any type can be made an array by adding `[]`.

```
tags   TEXT[],
marks  INTEGER[]

SELECT * FROM student WHERE 'Python' = ANY(subjects);
```

9. Enum

```
CREATE TYPE status AS ENUM ('active', 'inactive', 'pending');
```

10. Special Types

Category	Types
Network	INET, CIDR, MACADDR
Geometric	POINT, LINE, CIRCLE, POLYGON
Range	INT4RANGE, INT8RANGE, NUMRANGE, DATERANGE, TSRANGE, TSTZRANGE
Full-text search	TSVECTOR, TSQUERY
XML	XML
Money	MONEY (avoid, use NUMERIC)

Type Casting

```
SELECT '123'::INTEGER;
SELECT CAST('2026-10-28' AS DATE);
SELECT NOW()::DATE;
```

Quick Rules of Thumb

Situation	Choose
Money or exact decimals	`NUMERIC(10,2)`
Primary key	`GENERATED ALWAYS AS IDENTITY` or `UUID`
Names, emails, free text	`TEXT` or `VARCHAR(n)`
Date and time of events	`TIMESTAMPTZ`
Flexible or semi-structured data	`JSONB`
Yes/No flags	`BOOLEAN`
Fixed set of values	`ENUM` or a lookup table
Avoid	`MONEY`, `CHAR(n)`, `FLOAT` for currency

Example Table

```
CREATE TABLE student (
    student_id  INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY KEY,
    name        VARCHAR(100) NOT NULL,
    email       TEXT UNIQUE,
    dob         DATE,
    fees_paid   NUMERIC(10,2) DEFAULT 0.00,
    is_active   BOOLEAN DEFAULT TRUE,
    subjects    TEXT[],
    extra_info  JSONB,
    created_at  TIMESTAMPTZ DEFAULT NOW()
);
```